

MARKING REPORT — CHEMISTRY EXAM-03

Chapter 6: Stoichiometry | Seemab | 2026-05-12 | Corrected Rubric: 65 Marks

57.5 / 65

Section A: 12/12 | Section B: 32/33 | Section C: 13.5/20

88.5%

A+

Section A — MCQs

12/12

100% Perfect

Section B — Short Qs

32/33

97.0%

Section C — Long Qs

13.5/20

67.5%

Grand Total

57.5/65

88.5%

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question (short)	Correct	Seemab	Result	Mark
1	Empirical formula of glucose $C_6H_{12}O_6$	B — CH_2O	B	✓	1
2	Molecular mass of naphthalene $C_{10}H_8$	C — 128 amu	C	✓	1
3	Formula mass of $Mg(OH)_2$	C — 58 amu	C	✓	1
4	Particles in one mole	B — 6.022×10^{23}	B	✓	1
5	Moles of O_2 in 16 g	B — 0.5 mol	B	✓	1
6	Mass of 9.05 mol ozone O_3	C — 434.4 g	C	✓	1
7	Atoms in 1.25 mol Zn	B — 7.53×10^{23}	B	✓	1
8	Formula of aluminium oxide	B — Al_2O_3	B	✓	1
9	Same empirical and molecular formula	C — H_2O	C	✓	1
10	Mass of 4 mol H_2	A — 8.064 g	A	✓	1
11	Balancing by changing ____	C — coefficients	C	✓	1
12	Ions not participating — called ____	B — spectator ions	B	✓	1

Section A Total: 12 / 12 (10th consecutive perfect MCQ score)

SECTION B — Short Questions (11 × 3 = 33 Marks) — All 11 Attempted

Q.2(i) — MAIN: Define empirical and molecular formula with examples

3 / 3

Variant attempted: Main question

Empirical formula defined correctly as the simplest integer ratio of atoms; example HO for hydrogen peroxide given. Molecular formula defined as the expression specifying the actual number of atoms in each element in one molecule; example H_2O_2 given. Both definitions precise, both examples correct and matched.

Q.2(ii) — MAIN: Molecular mass of glucose $C_6H_{12}O_6$

3 / 3

Variant attempted: Main question

Correctly wrote: $6(12) + 12(1.008) + 6(16) = 72 + 12.096 + 96 = 180.096$ amu. All steps shown, formula identification, working, and final answer all correct.

Q.2(iii) — OR: Differentiate empirical vs molecular formula; two quantities needed

3 / 3

Variant attempted: OR

Clean two-column table comparing definitions (simplest integer ratio vs actual number of atoms), alternative names (experimental/simplest formula vs calculated formula), and examples (HO vs H_2O_2). The two quantities needed to derive molecular formula from empirical formula correctly identified as molecular mass and formula mass.

Q.2(iv) — MAIN: Mass of 0.25 mol CO₂**3 / 3***Variant attempted: Main question*

Molar mass of CO₂ correctly calculated as 1(12)+2(16)=44 g/mol. Formula mass = moles × molar mass applied correctly: 0.25 × 44 = 11 g. Minor: result labelled "11 g/mol" — the unit should be "g" only, but the numerical answer is unambiguous and correct in context.

Q.2(v) — MAIN: Molecules in 0.5 mol CH₄**3 / 3***Variant attempted: Main question*

Formula correctly stated (molecules = moles × N_A). Calculation: 0.5 × 6.022×10²³ = 3.011×10²³ molecules of methane. All three marks earned.

Q.2(vi) — MAIN: Chemical formulas of NaCl, Mg₃N₂, K₂SO₄**3 / 3***Variant attempted: Main question*

(a) Na⁺, Cl⁻ → NaCl: charge balance 1:1 shown, formula correct. (b) Mg²⁺, N³⁻ → Mg₃N₂: cross-charge rule applied, 3×2=6=2×3, formula correct. (c) K⁺, SO₄²⁻ → K₂SO₄: 2×1=1×2=2, formula correct. All three formulas earned full marks.

Q.2(vii) — OR: Balance 2Na(s) + 2H₂O(l) → 2NaOH(aq) + H₂(g)**3 / 3***Variant attempted: OR*

Unbalanced equation written with all state symbols. Atom counts shown for both sides; correct identification that H is unbalanced in the raw equation. Balanced equation 2Na(s) + 2H₂O(l) → 2NaOH(aq) + H₂(g) written correctly with state symbols. All steps present.

Q.2(viii) — OR: Number of atoms in 0.2 mol Al**3 / 3***Variant attempted: OR*

Moles=0.2 stated, N_A=6.022×10²³ noted. Correct formula applied: atoms = moles × N_A = 0.2 × 6.022×10²³ = 1.2044×10²³ atoms. (Note: she initially wrote "molar mass of Al" in her heading but struck it out — the calculation itself is entirely correct.)

Q.2(ix) — OR: Define ionic equation; ionic equation for HCl + NaOH**2 / 3***Variant attempted: OR*

Definition of ionic equation is correct (chemical equation where dissolved substance written using individual ions, spectator ions removed by net ionic equation).

–1 mark: All ion charges missing throughout the expanded and net ionic equations. HCl dissociates to H⁺(aq) + Cl⁻(aq), written as H + Cl without charges. NaOH dissociates to Na⁺(aq) + OH⁻(aq), written as Na + OH. The net ionic H + OH → H₂O should be H⁺(aq) + OH⁻(aq) → H₂O(l). No (aq) state symbols appear on any ion. An ionic equation without charges is not an ionic equation — it is a symbol equation. This is the same notation gap flagged on the previous marking.

Q.2(x) — OR: Mass of 1.2 mol K and 0.25 mol H₂O**3 / 3***Variant attempted: OR*

(a) K: molar mass=39 g/mol, mass=1.2×39=46.8 g — correct. (b) H₂O (steam): molar mass=2(1.008)+16=18.016 g/mol, mass=0.25×18.016=4.504 g — correct. Formula mass = moles × molar mass applied correctly for both parts.

Q.2(xi) — OR: Caffeine C₈H₁₀N₄O₂ — empirical formula and molecular mass**3 / 3***Variant attempted: OR*

(a) Ratio C:H:N:O = 8:10:4:2; GCD=2; empirical formula = C₄H₅N₂O — correct. (b) Molecular mass: 8(12)+10(1)+4(14)+2(16) = 96+10+56+32 = 194 amu — correct.

Section B Total: 32 / 33**SECTION C — Long Questions (4 × 5 = 20 Marks) — All 4 Attempted**

Q.3(i) — OR: Empirical/molecular definitions + Aspirin, Vinegar, TNT**4 / 5***Variant attempted: OR*

Part (a) [2/2]: Definitions of empirical formula (simplest integer ratio of atoms) and molecular formula (actual number of atoms per molecule) both correct. Examples: Benzene C_6H_6 ($6:6 \rightarrow 1:1 = CH$) ✓; Hydrogen peroxide H_2O_2 ($2:2 \rightarrow 1:1 = HO$) ✓. Full 2/2.

–0.5 mark: Aspirin $C_9H_8O_4$ — empirical formula written as $C_4H_2O_4$, which is incorrect. The ratio 9:8:4 has $GCD=1$, so the formula is already in its simplest form; the empirical formula is $C_9H_8O_4$ (same as molecular). A divisor of 2 gives 4.5, which is not a whole number — the attempted simplification is invalid. Molecular mass (180.064 amu with $H=1.008$, ~ 180 amu) is correct.

–0.5 mark: Vinegar CH_3COOH — empirical formula written as " CH_3COOH " (the molecular formula itself), not simplified. The correct empirical formula is CH_2O ($C_2H_4O_2$ reduces by $GCD=2$). Molecular mass (~ 60 amu) is correct.

TNT $C_7H_5N_3O_6$ [1/1]: Empirical formula $C_7H_5N_3O_6$ correct ($GCD=1$). Molecular mass $7(12)+5(1.008)+3(14)+6(16) = 84+5.04+42+96 = 227.04$ amu — correct.

Q.3(ii) — OR: Moles of He, C, S; mass of H_2 and H_2SO_4 **4 / 5***Variant attempted: OR*

Part (a)(i) [1/1]: He: mass=2.4 g, molar mass=4 g/mol, moles= $2.4/4=0.6$ mol ✓

–1 mark: Carbon (250 mg): Seemab used molar mass = 14 instead of 12. Carbon (C) has molar mass 12 g/mol; 14 g/mol is the molar mass of nitrogen (N). This gives moles= $0.25/14=0.0178$ mol instead of the correct $0.25/12=0.0208$ mol. The conversion 250 mg $\rightarrow$ 0.25 g is shown correctly. Error: C=12 confused with N=14 — the same elemental molar mass confusion noted in the previous marking report.

Part (a)(iii) [1/1]: S: mass=40 g, molar mass=32 g/mol, moles= $40/32=1.25$ mol ✓

Part (b)(i) [1/1]: H_2 : moles=75, molar mass= $2(1.008)=2.016$ g/mol, mass= $75 \times 2.016=151.2$ g ✓ (written as 151.29 — acceptable rounding).

Part (b)(ii) [1/1]: H_2SO_4 : moles=0.15, molar mass= $2(1.008)+32+4(16)=98.016$ g/mol, mass= $0.15 \times 98.016=14.7024$ g ✓

Q.3(iii) — OR: Ionic equation definition + three net ionic equations**2.5 / 5***Variant attempted: OR*

Part (a) [2/2]: Ionic equation defined correctly (chemical equation where dissolved substances written as individual ions). Spectator ions mentioned and concept of removal by net ionic equation stated — both marks earned.

–1 mark (Reaction i): Net ionic for $Mg + H_2SO_4$ written as $Mg + H_2 \rightarrow MgH_2$. This is fundamentally wrong. $H_2SO_4(aq)$ dissociates to $2H^+(aq) + SO_4^{2-}(aq)$. After removing spectator SO_4^{2-} , the correct net ionic is $Mg(s) + 2H^+(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$. Seemab has treated H_2 gas (product) as H_2 (reactant) and bonded it with Mg to form MgH_2 — a fictional compound in this context. All ion charges are also absent from the expanded ionic step.

–0.5 mark (Reaction ii): $AgNO_3 + NaCl \rightarrow AgCl(s) + NaNO_3$. Correct spectators identified (Na^+ and NO_3^-). Net ionic correctly identifies $Ag + Cl \rightarrow AgCl$ (species correct), but written without charges (should be $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$). Award 0.5/1 for correct spectator removal with wrong notation.

–1 mark (Reaction iii): $Zn + 2HCl \rightarrow ZnCl_2 + H_2$. Net ionic written as $ZnH + H_2 \rightarrow ZnH_2$. This repeats the same fundamental error as reaction i — treating H_2 gas as a reactant that bonds with the metal. Correct net ionic is $Zn(s) + 2H^+(aq) \rightarrow Zn^{2+}(aq) + H_2(g)$. All charges absent.

Q.3(iv) — MAIN: K + H_2O reaction — balance, moles, atoms**3 / 5***Variant attempted: Main question*

Part (i) Balance [1/1]: Atom counting shown correctly on both sides. Balanced equation: $2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$ — correct with state symbols.

Part (ii) Moles of K [2/2]: mass=0.5 g, molar mass of K=39 g/mol. Moles = $0.5 \div 39 = 0.0128$ mol — correct. (Unit label written as "0.0128 mol/g" — minor notation slip, numerical answer correct.)

–2 marks: Part (iii) — Number of K atoms. Written as: atoms = molar mass $\times N_A = 39 \times 6.022 \times 10^{23} = 2.34 \times 10^{25}$. This is categorically wrong. The formula is atoms = moles $\times N_A$. The value to multiply by N_A is the number of moles (0.0128), NOT the molar mass (39). Correct answer: $0.0128 \times 6.022 \times 10^{23} = 7.71 \times 10^{21}$ atoms. This is the third consecutive Chemistry exam where the three-step chain mass $\rightarrow$ moles ($\div M$) $\rightarrow$ atoms ($\times N_A$) breaks at the final step. The molar mass is already used in step (ii); it cannot be reused in step (iii) as the multiplier for N_A .

Strengths and Weaknesses Summary

Strengths demonstrated in this attempt

- **Perfect MCQ score (12/12):** 10th consecutive perfect MCQ score — every single MCQ across the last 10 sittings answered correctly. This is an exceptional record.
- **Mass-mole-mass conversions in Section B:** mass = moles $\times$ M and moles = mass $\div$ M applied consistently and correctly in Q2(iv), Q2(x), Q2(xi) and Q3(ii)(b). These are now automatic.
- **Empirical formula simplification (when GCD > 1):** Caffeine $C_8H_{10}N_4O_2 \rightarrow C_4H_5N_2O$ correctly simplified in Q2(xi). TNT correctly identified as already simplest.
- **Balancing chemical equations:** Q2(vii) and Q3(iv)(i) both balanced perfectly with all state symbols.
- **Chemical formula derivation from ions:** Q2(vi) — all three formulas ($NaCl$, Mg_3N_2 , K_2SO_4) derived correctly using the cross-charge method with workings shown.
- **Particle count formula (Section B):** Q2(v) and Q2(viii) both correctly used particles = moles $\times N_A$. The error only appears when the question requires going back through the moles step.

Recurring errors — targeted revision required

- **moles $\times N_A$, not molar mass $\times N_A$ (Q3(iv)(iii) –2 marks):** This is the third chemistry exam where the atom-count step uses the wrong quantity. The chain has three steps: mass $\div$ M = moles, then moles $\times N_A$ = particles. Once molar mass (M) is used in step 1, it does not appear again. Drill: write "moles $\times N_A$ " at the top of every particle-count calculation, then substitute the moles answer from the previous sub-part.
- **Ion charges in ionic equations (Q2(ix) –1 mark, Q3(iii) –2 marks):** Every ion in an expanded or net ionic equation must carry its charge superscript (H^+ , Cl^- , Mg^{2+} , SO_4^{2-}) and (aq) state symbol. Writing " $H + Cl$ " is a symbol equation, not an ionic equation. Three separate questions across two exams now show this gap.
- **Net ionic products in acid-metal reactions (Q3(iii)(i) and (iii) –2 marks):** When an acid reacts with a metal, the net ionic is Metal + $H^+(aq) \rightarrow Metal^{n+}(aq) + H_2(g)$. H_2 is a product (gas), not a reactant. MgH_2 and ZnH_2 do not form in aqueous acid. Writing " $Mg + H_2 \rightarrow MgH_2$ " shows the H_2 on the wrong side of the arrow.
- **C = 12, not 14 (Q3(ii)(a)(ii) –1 mark):** Carbon molar mass is 12 g/mol; Nitrogen is 14 g/mol. These are adjacent on the periodic table and one position apart in value. Make a flash card: C=12 (Z=6), N=14 (Z=7).
- **Empirical formula simplification when GCD = 1 (Q3(i)(b) –0.5):** Aspirin $C_9H_8O_4$ — dividing by 2 gives non-integer subscripts (4.5), which is invalid. Always check: does dividing produce whole numbers? If not, the formula is already empirical.
- **Vinegar empirical formula (Q3(i)(b) –0.5):** CH_3COOH is the structural/molecular formula. Writing it as the "empirical formula" misses the step of expanding to $C_2H_4O_2$ and dividing by GCD=2 to get CH_2O .

Performance Comparison — Chemistry Exams

Exam	Date	Chapters	Total	%	MCQ	Sec B	Sec C
Chem exam-01	2026-04-12	Ch 1-2	64/65	98.5%	12/12	31/33	21/20*
Chem exam-02	2026-04-17	Ch 3-5	59.5/65	91.5%	12/12	29/33	18.5/20
Chem exam-03 (corrected)	2026-05-12	Ch 6	57.5/65	88.5%	12/12	32/33	13.5/20

* exam-01 Sec C figure is approximate from prior records. Chemistry average across 3 exams: 92.8%.

Note: exam-03 originally marked at 44.5/46 (96.7%) under a "best 8 of 11 / best 2 of 4" rubric. The corrected 65-mark rubric (all 11 + all 4) reveals Section C to be significantly weaker (67.5%) than Section B (97.0%). This masked pattern now visible for targeted revision.